

AWS EC2 — Deploying Twenty CRM

1. Objective

This document records the steps taken to launch and configure an AWS EC2 instance, connect to it, install Docker, and deploy the Twenty CRM application using its provided Docker configuration.

2. Environment Summary

Component	Configuration
Operating System (AMI)	Ubuntu Server 22.04 LTS
Connection Method	SSH via Git Bash
Container Runtime	Docker
Application Deployed	Twenty CRM (open-source CRM)
Deployment Method	Cloned application repository containing a Dockerfile / Docker Compose configuration

3. Deployment Procedure

3.1 Launch the EC2 Instance

1. Open the AWS Console and navigate to the EC2 service.
2. Launch a new EC2 instance using the Ubuntu 22.04 LTS AMI.
3. Selected t3.small instance type
4. Create or select an existing key pair for SSH access and download the .pem key file.
5. Configure the security group to allow inbound traffic on the required ports, including SSH (port 22) and the application port used by Twenty CRM.
6. Review the configuration and launch the instance.

3.2 Connect to the Instance

Connect to the instance from the local machine using SSH via Git Bash.

```
chmod 400 twentycrmkey.pem
ssh -i twentycrmkey.pem ubuntu@52.54.85.180 (server public ip)
```

3.3 Install Docker

Once connected, Docker is installed on the Ubuntu instance to support running the containerized Twenty CRM application.

```
sudo apt update
sudo apt install -y docker.io docker-compose-plugin
sudo systemctl enable --now docker
sudo usermod -aG docker $USER
```

Verify the installation:

```
docker --version
docker compose version
```

3.4 Clone the Twenty CRM Repository

Clone the Twenty CRM of branch for task5 where dockerfile exist in the repository onto the instance. The repository includes the Dockerfile / Docker Compose configuration required to run the application.

```
git clone https://github.com/PearlThoughts-Intern-DevOps/devops-crm-project
cd devops-crm-project
```

3.5 Deploy and Run the Application

Build and start the application using Docker:

```
docker compose up -d
```

Confirm that the containers are running:

```
docker ps
```

The Twenty CRM application is then accessed using the instance's public IP and the configured application port.

4. Key EC2 Concepts Used

- **AMI (Amazon Machine Image):** The Ubuntu 22.04 template used to launch the EC2 instance.
- **Instance Type:** Determines the compute, memory, and network capacity.
- **Key Pair:** The public/private key used for secure SSH authentication to the instance.
- **Security Group:** Acts as a virtual firewall controlling inbound and outbound traffic, including SSH and application ports.
- **Elastic/Public IP:** Used to reach the instance and the running application from outside AWS.

5. Outcome

The EC2 instance was successfully launched and configured, connected to via SSH, and Docker was installed. The Twenty CRM repository was cloned and deployed using its Docker configuration, resulting in the application running successfully on the instance.